Atlanta, GA 30329 | Resume | 11/7/26
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DEREK EUBANKS PE, HFDP, HBDP, CPD
SENIOR MECHANICAL PE | GEORGIA TECH M.S. COMPUTER SCIENCE (AI) CANDIDATE | AI-ASSISTED POLYGLOT PROGRAMMER

PROFESSIONAL SUMMARY

Licensed Mechanical Engineer (PE – CA, GA, TN) with 13+ years of experience leading HVAC, plumbing, medical gas, and hydronic design for healthcare, hyperscale data centers, and advanced manufacturing facilities. Former Mechanical Discipline Director with expertise in central utility plants, chilled- and heating-hot-water systems, energy recovery, critical-space ventilation, HVAC controls, and multidisciplinary coordination. Advanced in Revit-based BIM design, equipment selection, engineering calculations, construction documentation, and technical QA/QC; holds HFDP, HBDP, and CPD certifications. Currently pursuing an M.S. in Computer Science at Georgia Tech, specializing in Artificial Intelligence. Applies Python, C# (Revit API), VBA, Java, machine learning, and AI-assisted programming to develop custom Revit families and automation tools, streamline workflows, improve accuracy, and support data-driven building-system optimization.

SKILLS & SOFTWARE

- Autodesk Revit Expert
- Bluebeam Revu
- Microsoft Office Suite
- Python, C# (Revit API), VBA, Java
- D3.js, Matplotlib, Seaborn, Pandas, NumPy, SciPy, Scikit-learn
- Trane Trace 3D Plus; Carrier HAP
- High-performance Mission-Critical and Healthcare HVAC & Plumbing & Medical Gas Design
- Hydronic waterside plant design
- QA/QC for HVAC, plumbing, and medical gas drawing markups
- ML, AI, Ensemble Learning & Predictive Analytics
- HVAC Controls, BAS Sequences of Operation, Control Schematics, Points Lists, & Hydronic Flow Diagrams
- AI-Assisted Software Programming

EXPERIENCE

MECHANICAL ENGINEER PE III • ENFRA • JUNE 2026 – PRESENT; REMOTE

- Lead HVAC design and multidisciplinary coordination for a new 8-story healthcare facility, including VAV energy-recovery AHUs, chilled-water and heating-hot-water systems, associated ductwork and hydronic distribution.
- Engineer specialized HVAC pressure-controlled systems with terminal air valves for Airborne Infection Isolation (AII), Protective Environment (PE), CATH/IR rooms, and class III imaging rooms - including high-plume exhaust, load calculations, airflow calculations, and positive/negative pressure relationships in accordance with ASHRAE 170, 90.1, 62.1, FGI, IMC, IPC, NFPA, and project-specific design criteria.
- Research and develop AI-assisted and Revit-based QA/QC workflows to automate equipment, schedule, model, and drawing consistency checks.
- Develop building-wide chilled-water and heating-hot-water hydronic flow diagrams, HVAC control schematics, sequences of operation (SOO), and points lists.

SENIOR MECHANICAL ENGINEER • SPACEX • JULY 2025 – JAN 2026; ON-SITE

- Engineered variable-primary-flow heat-pump chiller plants, HVAC systems, energy recovery ventilation/exhaust systems, and hydronic distribution for advanced manufacturing facilities using Trace 3D Plus and Revit.
- Developed complete 3D Revit MEP models, equipment layouts, hydronic piping, HVAC controls schematics, points lists, and sequences of operation (SOO) from field-verified existing conditions.
- Delivered complete demolition and new-construction mechanical drawing packages using Revit—from laser-verified field existing conditions to permit-ready documentation—in ≤ one week.

SENIOR ASSOCIATE/SENIOR MECHANICAL ENGINEER • SYSKA HENNESSEY GROUP • JAN 2024 – MARCH 2025; ON-SITE

- Led mechanical design for a 48 MW Tier III hyperscale data center, developing N+1 chilled-water and HVAC infrastructure supporting data halls, CRAH galleries, mechanical pump rooms, and critical electrical spaces.
- Engineered and coordinated a ~21,800 GPM chilled-water system, 40+ air-cooled chillers, CRAH galleries, and three mechanical hydronic pump rooms in Revit. Directed and developed a PipeFlo model for the system.
- Directed mechanical design development, equipment selection, load analysis, and interdisciplinary BIM coordination through permit-level construction documentation.

SENIOR MECHANICAL ENGINEER • JACOBS • OCT 2022 – DEC 2023; REMOTE

- Designed HVAC and hydronic systems for Tesla and Rivian advanced-manufacturing facilities, performing load calculation analysis and full 3D Revit coordination.
- Engineered and modeled multiple 40,000+ CFM chilled-water AHUs, associated ductwork and piping, heat-pump systems, and low-dewpoint environmental-control system schematics, points lists, and SOOs.
- Coordinated multidisciplinary design development while delegating calculations and design tasks to junior engineers.

*Full portfolio, code, and project writeups available via GitHub and LinkedIn

ADDITIONAL ENGINEERING EXPERIENCE:

- **SENIOR MECHANICAL ENGINEER • MAZZETTI • APR 2020 – MAY 2023; REMOTE**
  - Designed HVAC, plumbing, and medical-gas systems for major healthcare facilities using Revit, Trace 3D Plus, and detailed psychrometric/hydronic analysis - improving energy efficiency, design accuracy, code compliance, and coordination.
  - Developed VBA-based engineering tools and detailed BAS controls schematics, points lists, and sequences of operation.
- **MECHANICAL DISCIPLINE DIRECTOR • GRAY AES • AUG 2018 – MAR 2020; ON-SITE**
  - Served as Mechanical EOR and discipline leader, directing engineers/designers, managing project assignments, and leading HVAC/plumbing design for healthcare and industrial clients.
  - Led proposals, project management, performance reviews, and mechanical discipline coordination.

EDUCATION:

- **MASTER OF SCIENCE IN COMPUTER SCIENCE • GEORGIA INSTITUTE OF TECHNOLOGY • 2025–2028**
  - Specialization: Artificial Intelligence
  - GPA: 3.67/4.0
  - Completed: CS 6603, CSE 6242, CS 6310
  - In Progress: CS 6601 - Artificial Intelligence (Fall 2026)
- **BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING • GEORGIA INSTITUTE OF TECHNOLOGY • 2007–2013**
  - Dean's List (2) semesters; Merit list (2) semesters
  - Math GPA: 3.68/4.0 (7 courses); Computer Science GPA: 3.5/4.0 (2 courses)
  - HOPE scholarship & Federal PELL Grant recipient for maintaining >= 3.0 GPA

GRADUATE AI, ML, & COMPUTATIONAL PROJECTS:

- Adversarial Search Agent | Python | CS 6601 Artificial Intelligence
  - Developed a game-playing agent using Minimax, Alpha-Beta Pruning, Iterative Deepening, heuristic evaluation, and depth-limited adversarial search.
  - Implemented recursive MAX/MIN logic and runtime-aware search optimization; earned 100/100.
- Graph & Heuristic Search Algorithms | Python | CS 6601 Artificial Intelligence
  - Implemented BFS, UCS, A*, Bidirectional UCS, Bidirectional A*, and Tridirectional Search for weighted graph traversal and optimal-path discovery.
  - Applied priority queues, admissible heuristics, frontier management, and path-cost optimization.
- Decision Trees & Ensemble Boosting | Python | CS 6601 Artificial Intelligence
  - Implemented supervised classification algorithms and ensemble boosting techniques, achieving 98.10% test accuracy.
  - Evaluated predictive performance, generalization, and execution efficiency using multiple performance metrics.
  - Developed decision-tree classifiers using Gini impurity, information gain, recursive splitting, and weighted ens. learning.
- Airline Delay Root Cause Explorer | Python / Scikit-learn | CSE 6242 Data & Visual Analytics
  - Built an end-to-end ML pipeline using 400k–800k+ BTS/NOAA records, including feature engineering, preprocessing, modeling, and evaluation.
  - Compared Logistic Regression, Random Forest, and Gradient Boosting models; achieved ROC-AUC ~0.70 and performed feature-importance analysis.
- Revit API Plumbing Engineering Automation | C#
  - Developed custom Revit API functionality integrating plumbing fixture-unit conversions to flow rate, hydraulic calculations, and automated pipe sizing with a single click event to improve design consistency and reduce manual effort.
  - Created custom parametric Revit plumbing families incorporating pipe connectors, fixture-unit parameters, calculated flow rates, and conditional formulas – all per IPC - to integrate with the C# add-in and support automated domestic water system sizing.

LICENSES & CERTIFICATIONS:

- Professional Engineer (PE Licenses): California (#42960), Georgia (#PE042696), Tennessee (#130418)
- MyNCEES PE-verified engineering employment history since 2013 – eligible to pursue comity licensure in other states.
- ASHRAE Healthcare Facility Design Professional (HFDP)
- ASHRAE High-Performance Building Design Professional (HBDP)
- ASPE Certified in Plumbing Design (CPD)
- Autodesk Certified Professional in Revit for Mechanical Design

*Full portfolio, code, and project writeups available via GitHub and LinkedIn